MongoDB
Pour aller plus loin

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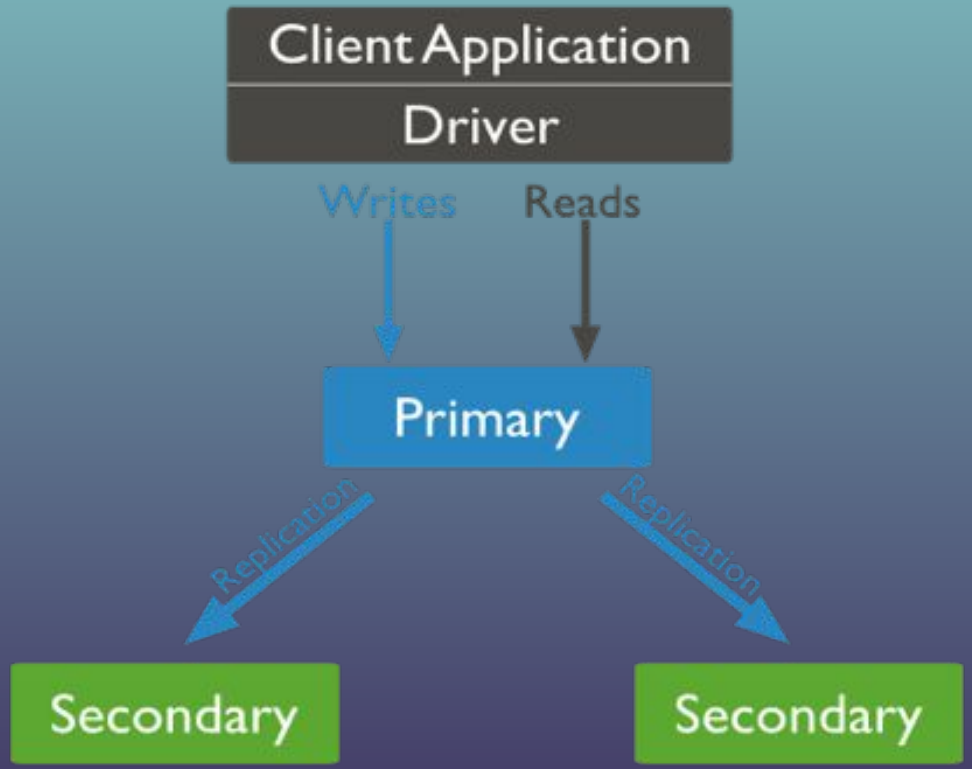
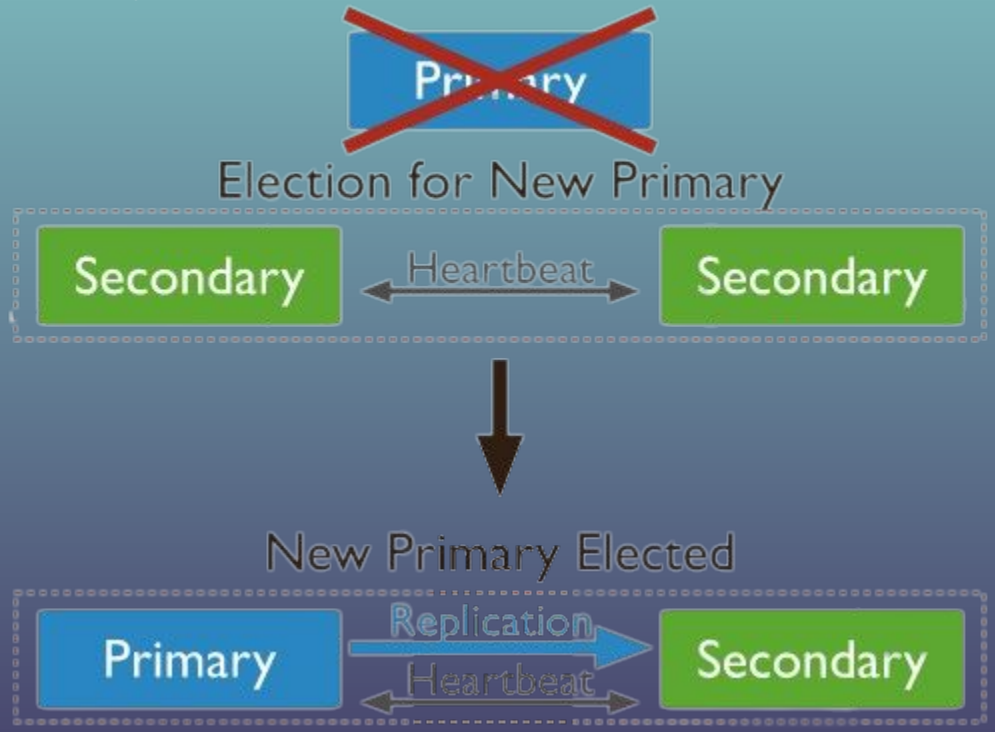
Réplication de données



Concepts :

- Au moins 3 serveurs
- Répartition des taches
- Copie constante entre Primary et Secondary
- Facilité de sauvegarde
- Gestion de données massives

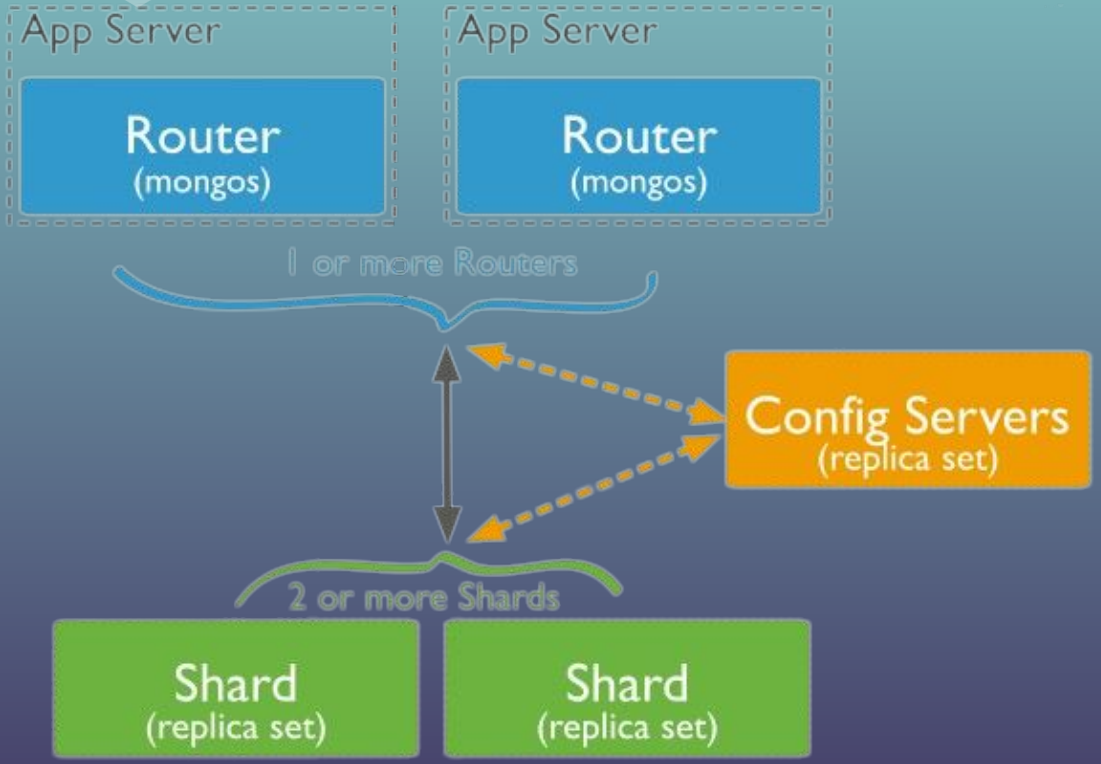
Réplication de données



Tolérance aux pannes :

- Election d'un nouveau primary
- Prise de relais automatique
- Disponibilité constante des données

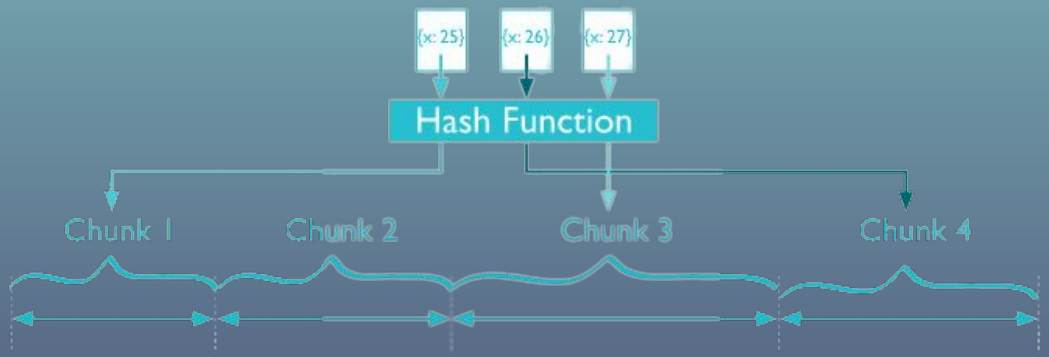
Sharding



Concepts :

- Diviser pour mieux régner
- Config servers -> Router -> Shard
- Amélioration des performances

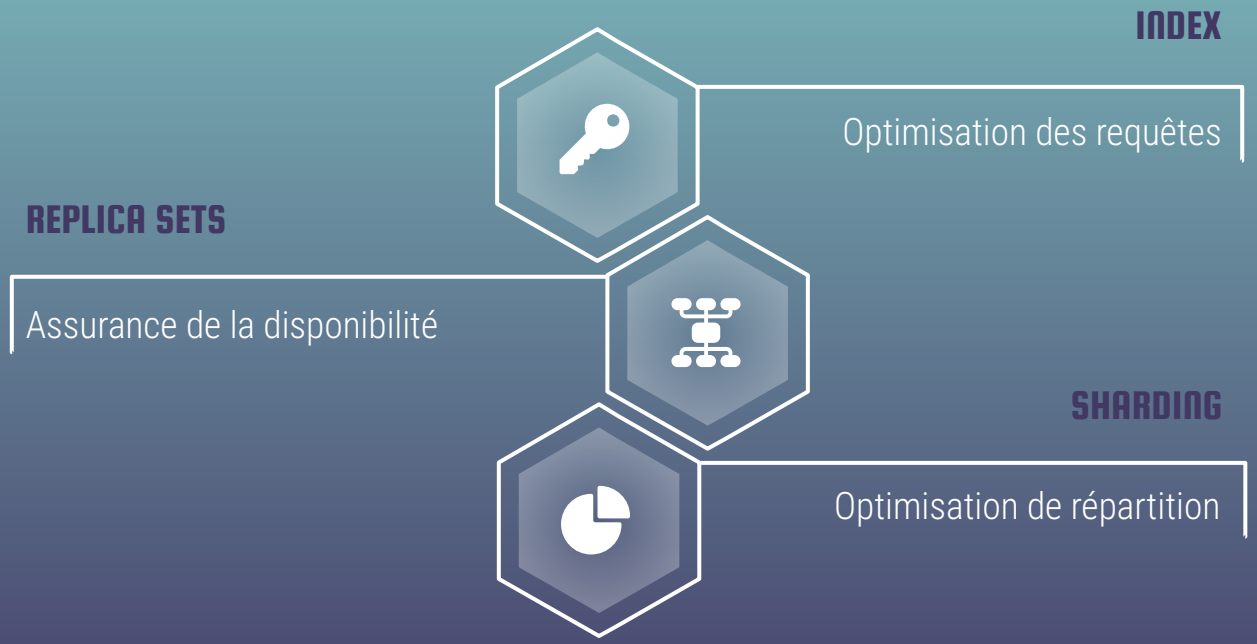
Sharding



Evolution horizontale :

- Données massives
- Evolutivité
- Répartition de la charge
- Capacité de stockage
- Isolation des données

Les clés de la réussite en MongoDB



Application en langage back – PHP

Extension officielle “mongodb”

```php
<?php
$manager = new MongoDB\Driver\Manager("mongodb://localhost:27017");
$collection = new MongoDB\Collection($manager, "testdb", "testcollection");

$document = [
    "name" => "John",
    "age" => 30,
    "city" => "New York"
];

$insertResult = $collection->insertOne($document);
echo "Inserted ID: " . $insertResult->getInsertedId();
?>
```

Application en langage back – Node.js

Module mongodb-driver

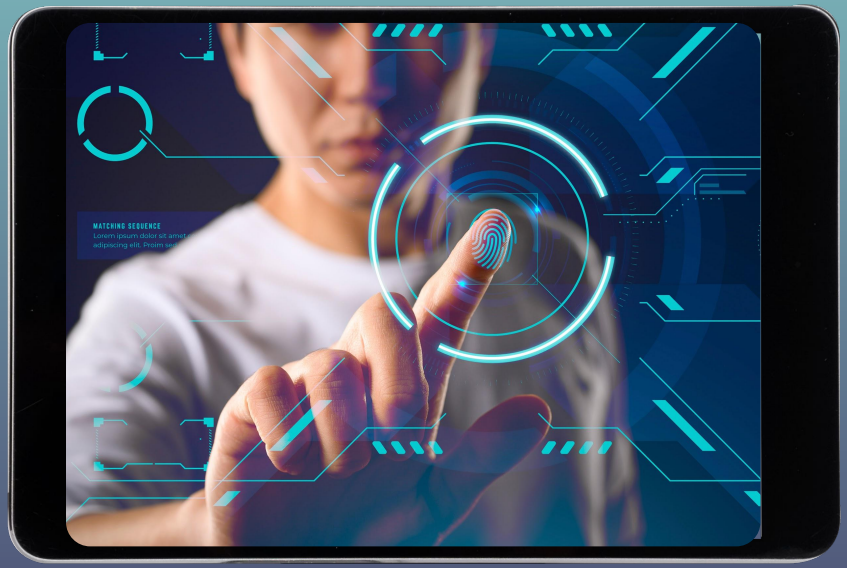
```javascript
const { MongoClient } = require("mongodb");

const uri = "mongodb://localhost:27017";
const client = new MongoClient(uri);

async function run() {
    try {
        await client.connect();

        const database = client.db("testdb");
        const collection = database.collection("testcollection");

        const document = {
            name: "Alice",
            age: 28,
            city: "London"
        };

        const insertResult = await collection.insertOne(document);
        console.log("Inserted ID:", insertResult.insertedId);
    } finally {
        await client.close();
    }
}

run().catch(console.error);
```

Application en langage back – Java

Driver officiel MongoDB Java

```java
import com.mongodb.MongoClient;
import com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import org.bson.Document;

public class MongoDBExample {
public static void main(String[] args) {
MongoClient mongoClient = new MongoClient("localhost", 27017);
MongoDatabase database = mongoClient.getDatabase("testdb");
MongoCollection<Document> collection = database.getCollection("testcol");

Document document = new Document("name", "Michael")
.append("age", 25)
.append("city", "Los Angeles");

collection.insertOne(document);
System.out.println("Inserted ID: " + document.get("_id"));

mongoClient.close();
}
}
```

Merci pour votre attention



Bon courage pour la suite

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